Act Through the Latent: Toward a Learned Retarget-to-Track Interface for Humanoid Tracking

Anonymous ICRA Submission

***Abstract*—The dominant humanoid motion pipeline is two-stage: retarget human motion to the robot, then train a tracking policy on the result. The interface between the stages is a stack of robot-space joint targets—a representation chosen for convenience, not for control. We take a first step toward a *learned* interface. We distill a per-frame IK retargeter into one multi-embodiment network with a shared retargeting latent z_{ret} (128-d, five humanoids), and audit that latent with pre-specified probes: it is instance-aligned across embodiments (window retrieval 57–83% top-1 at 1.02% chance—the low end is the morphological outlier; CKA 0.86–0.92), with motion category not linearly decodable (0.151 vs 0.125 chance) though nonlinearly organized, is embodiment-aligned but not invariant (0.28 linear probe vs 0.91 nonlinear attacker), physically impoverished (contact F1 0.088 without supervision), and inert when concatenated beside an explicit reference, a null that an independent ablation confirms. On the control side we show a separate 64-d command code z_{cmd}, produced by a goal-conditioned residual-to-prior CVAE and DAgger-distilled from an explicit-command RL teacher, can be the actor's *only* motion input at 0.955 ± 0.008 ten-second completion—*matching* its teacher (0.954 at equal evaluation precision; 3 seeds, 1,024 rollouts each). Feeding the prior the frozen retargeting latent *instead of* any robot-space reference reaches 0.656 ± 0.044 where a goal-free prior collapses to 0.001—but a pre-specified control we ran at reviewers' insistence overturned our own favored reading: a pure *clock* (a fixed random projection of frame-index sinusoids through the identical fetch path) reaches 0.754 ± 0.010, so on single-clip cyclic motion the latent's deployable value does not exceed—indeed falls short of—phase information alone. We report this as the third result our falsification apparatus killed, and as the null model single-clip act-through-latent claims must beat. The retargeting network also absorbs a second, interaction-rich teacher across three human skeletons, cutting interaction-clip articulation error 3.1–$5.2\times$ at $+1.05$ cm on locomotion (preliminary: ground-truth root). We release a pre-specified negative-results ledger and two measurement defects—one in a widely used simulator framework whose repair cut joint tracking error by 46%, and one in our own trainer whose repair first overturned our own logged negative on the frozen latent and whose clock control then retired the favorable reading: both reversals gated, both reported.**

I. Introduction

Nearly every humanoid that has learned to move from human motion has done so in two stages: a retargeter converts human motion into robot-space joint trajectories, and a reinforcement-learning tracking policy learns to follow them. The split is a good engineering decision—retargeters and trackers are developed, evaluated, and reused independently, and the recent generation of each is strong: interaction-preserving optimization on the retargeting side [2], physics-in-the-loop retargeting [1], and unified tracking controllers that follow thousands of clips with one network [4], [5]. But the *interface* between the stages has escaped scrutiny. What actually crosses the boundary is a per-frame stack of joint targets—a format, not a representation.

That format is lossy in three specific ways. It discards everything the retargeter knew that is not a joint angle: its contact reasoning, its uncertainty, and the cross-embodiment structure it learned when one network serves several robots. It is evaluated by kinematic metrics (pose error, foot skate, penetration) that are blind to *trackability*—the GMR study [3] showed retargeting artifacts silently cap what downstream RL can learn, evidence that the interface transmits harm as readily as signal. And it is *dimensionally unshareable*: a 29-DoF joint-target stream cannot command a 23-DoF robot, so every multi-robot system pays the interface cost once per embodiment.

The obvious repair fails, and the failure is informative. Handing the tracking policy the retargeter's latent *alongside* its usual reference does nothing: in our experiments the policy ignores the latent (concatenation is null-to-harmful, -10 points at worst), and UniTracker's ablation reports the same mechanism independently—"when the actor receives the reference motion directly, the influence of the latent variable z vanishes" [6]. Training a latent by RL from scratch collapses [6], [15]. Remove the explicit reference entirely and the frozen latent does carry the policy—0.656 ± 0.044 completion as the sole goal source, against 0.001 goal-free—but the decisive control cuts deeper than the encouraging number: a pure clock, a fixed random projection of frame-index sinusoids fed through the identical pathway, reaches 0.754 ± 0.010. On a single cyclic clip, phase information alone outperforms the retargeting latent; whatever motion content the latent holds beyond phase is not deployable value on this instrument, and its phase is noisier than clean sinusoids. The latent is *dominated* twice over.

Our key insight is that the retargeting-to-tracking boundary should not be a representation the two stages agree on, but one the tracking policy learns to act through: a latent becomes useful for control exactly when it is the policy's only channel to the goal, and is ignored whenever a more informative explicit reference is available beside it. The sections that follow are the constructive and destructive halves of that claim. We build SNMR, a skeleton-agnostic retargeting network distilled from a classical IK teacher whose shared latent serves five humanoids, and we audit that latent with falsification-first probes before asking it to do anything (§IV). We then make the interface load-bearing (§V): a conditional VAE whose goal-conditioned prior compresses the reference into a 64-d command latent, trained by DAgger from an explicit-command RL teacher, with the reference

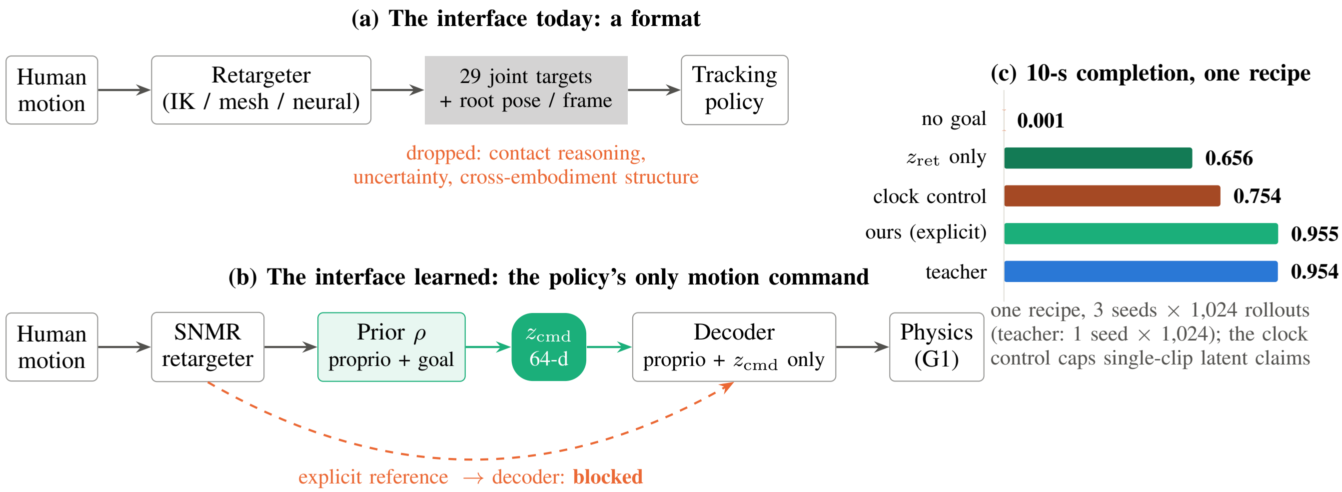

Fig. 1. **The retarget-to-track interface, learned (single-clip scope; see §VIII).** (a) The standard humanoid stack passes per-frame joint targets between retargeting and tracking, discarding the retargeter's contact reasoning, its uncertainty, and the cross-embodiment structure it learned. (b) We replace that interface with a learned 64-d command latent: the goal reaches the goal-conditioned prior but *never* the decoder, so z_{cmd} is the tracking policy's only motion command. (c) Under one training regime (3 training seeds, 1,024 rollouts per evaluation): the learned code with the explicit goal *matches* its teacher (0.955 vs 0.954); the frozen human-side retargeting latent alone reaches 0.656—but a phase-only clock control reaches 0.754, capping what the latent can claim on a single cyclic clip.

visible to the prior but never to the decoder. On a simulated Unitree G1 this interface *matches* its teacher—0.955 ± 0.008 vs 0.954 at equal evaluation precision (Fig. 1c); replacing the prior's robot-space goal with the frozen retargeting latent—no robot-space reference anywhere in the student—retains 0.656 ± 0.044 against a 0.001 goal-free control, but a phase-clock control at 0.754 caps that arm's interpretation at "timing, at most." The same latent then absorbs a second, interaction-rich teacher—terrain and object clips from a fundamentally different retargeter, arriving on different human skeletons—cutting interaction-clip error 3–5$\times$ at a measured $+1.05\,$cm locomotion cost (§VI).

Two findings we did not go looking for became contributions. First, a world-body indexing defect in a widely used MuJoCo-Warp training framework silently zeroed the primary body-position tracking reward in every one of our runs (upstream report in preparation); its repair cut joint tracking RMSE by 46% with no change to the learning algorithm and brought our tracker into an externally calibrated band (§VII). Second, rolling tracking policies out on their own references and recording the simulated states yields physically repaired supervision—2–3$\times$ less foot skate at a measured fidelity price—quantifying the data-side alternative to physics-in-the-loop retargeting.

Contributions.

- **A learned command interface for humanoid tracking, under an exclusivity contract.** A goal-conditioned, residual-to-prior CVAE whose 64-d latent is the policy's only motion input—0.955 ± 0.008 completion, *matching* its explicit-command teacher (0.954 at equal precision; 3 seeds), with the reference never visible to the decoder and the code measurably a *recoding*, not a copy (effective rank 14 of 64; the goal is less linearly decodable from z_{cmd} than from proprioception alone, held-out R^2 0.48 vs 0.62)—plus a documented failure anatomy (path consistency, goal conditioning, observation-layout leaks) for making such interfaces work.
- **The clock null model for act-through-latent claims.** A pre-specified control that overturned our own favored result: on single-clip cyclic tracking, a fixed random projection of frame-index sinusoids as the prior's only input reaches 0.754 ± 0.010 completion—*above* the frozen retargeting latent (0.656 ± 0.044) and 79% of teacher. Any claim that a learned latent "carries motion content" on periodic clips must beat this trivially-constructed bar; ours did not, and we say so.
- **A falsification-first audit of a shared motion latent.** Pre-specified probe families establishing what the latent carries—instance-aligned, embodiment-aligned but not invariant, physically impoverished, dominated beside an explicit reference—with the negative verdicts stated as measured results, one independently replicated and one *overturned by our own defect repair* (frozen-latent control value: from "inert" to two-thirds of teacher).
- **Skeleton-agnostic retargeting distillation, measured honestly.** A 1.5M-parameter network amortizing an IK teacher across five humanoids (limits by construction, 671 fps CPU), and a separate two-teacher G1 study absorbing interaction-rich clips across three human skeletons at a quantified locomotion cost (preliminary: articulation under ground-truth root)—with the sharing cost and the zero-shot-embodiment failure (5.2$\times$) measured rather than assumed.
- **A measurement-substrate defect, found, fixed, and released**, plus the pre-specified negative-results ledger—closed research lines with the evidence that closed them,

so the field can stop re-deriving them.

II. Related Work

Retargeting as preprocessing. Classical retargeting solves per-frame IK with hand-tuned correspondences [3]; interaction-mesh methods add hard contact constraints [2]; ReActor moves retargeting inside physics via bilevel RL, eliminating artifacts by construction [1]. All three families emit the same interface—robot joint trajectories—and are evaluated by kinematic quality plus, recently, downstream RL success [3], [2]. None asks whether the *representation* handed to the tracker should itself be learned. Neural feed-forward retargeters [12], [13], [14] amortize the optimization and, in AdaMorph's case, span many embodiments through a morphology-agnostic latent—but their latents are internal machinery, not command interfaces, and none is analyzed for what it carries.

Tracking and its command spaces. Modern trackers follow references specified as joint targets with body-pose terms [4], [5], [11]; HOVER unifies multiple command *modes* by distillation [10]; UniTracker compresses the reference into a CVAE latent and is the closest precedent for our command interface—single-embodiment, with no retargeter and no representation analysis [6]. Latent skill spaces for physics characters [7], [8], [9] establish the recipe elements we verify at robot scale: learned conditional priors, residual-to-prior encoders, distillation-not-RL. Behavioral foundation models [16] reach a promptable latent from the opposite entry point—unsupervised RL with no retargeting anywhere. To our knowledge no published system co-trains a *multi-embodiment retargeting* latent under the *control* objective.

Interface critiques. BeyondMimic names a "planning-control gap" [4]; the GMR study quantifies how retargeting quality caps tracking [3]. Both stop at the boundary we cross: neither changes what the boundary *is*.

III. The Two-Stage Interface

A source motion m_t enters a retargeter R; the standard interface hands the tracker π a reference $g_t = (q_t^{\mathrm{ref}}, \dot{q}_t^{\mathrm{ref}})$ of robot joint targets. We move the actor-visible reference behind a learned code: $z_{\mathrm{cmd}} = \mu_\rho(\mathrm{proprio}_t, g_t)$ (the current-frame command; a future window is future work), under an *exclusivity contract*: the goal is visible to the prior ρ and to the critic, but never to the decoder, so motion intent reaches the actor only through the 64-d code. We use "interface," not "bottleneck": the code is not dimensionally narrower than the goal (64-d each), and no rate term constrains the goal-to-code path—the claim is exclusivity and recoding, which we verify directly (the deployed code has effective rank 14 of 64, and the goal is *less* linearly decodable from it than from proprioception, held-out R^2 0.48 vs 0.62: the prior discards reference detail and emits a control-sufficient code rather than a copy). The prior's goal source is the experimental factor (§VI): the robot-space reference g_t (the actor-side interface), the source-side representation z_{ret} alone (the interface-replacement arm—no robot-space reference anywhere in the student), both, or neither (control).

The latent's source is SNMR: a graph-attention encoder over the source skeleton (per-node features, adjacency, global pooling—skeleton-agnostic by construction) maps motion to a per-frame retargeting latent $z_{\mathrm{ret}} \in \mathbb{R}^{128}$ (distinct from the control code z_{cmd}); embodiment-conditioned graph decoders (AdaLN on a 32-d embodiment code, $\tanh$ joint-limit heads) emit trajectories for any of five humanoids. Trained by distillation from a per-frame IK teacher [3] on 77 LAFAN1 clips $\times$ 5 robots (2.48M paired frames), it reaches 3.66 cm held-out MPJPE (CI $[3.46, 3.86]$; G1 specialist; 2.9–6.0 cm shared across all five robots) at 671 fps on CPU, with zero joint-limit violations by construction. Foot skate (0.29 vs teacher 0.05 m/s) is the known inherited gap and is treated as a measured limitation, not hidden.

IV. Auditing the Latent

Before asking the latent to carry control, we measured what it holds. Four pre-specified probe families, in increasing severity:

Content: instance-aligned, not linearly semantic. Cross-embodiment *window* retrieval hits 75–83% top-1 at 1.02% chance on four embodiments and 57% on the fifth (the morphological outlier); CKA between per-robot encodings is 0.86–0.92. A *linear* motion-category probe is near chance (0.151 vs 0.125), but low-dimensional embeddings show clear category clusters: the structure is nonlinearly organized, and we claim only that it is not linearly accessible—the same caveat we apply to the embodiment probe below.

Embodiment: aligned, not invariant. A linear probe recovers the source embodiment at only 0.28 accuracy, but a nonlinear attacker reaches 0.91—the classic gap between "not linearly decodable" and "absent." A height-normalization intervention moves the attacker only $\sim$1 pp, suggesting—though, being an out-of-distribution intervention, not proving—that body scale is a minor share of the leak.

Physics: impoverished. Foot-contact state is weakly present under pure distillation: linear F1 0.088 at full budget—at the always-predict-contact baseline for the label prevalence ($\approx$0.09), with AUROC 0.51–0.64 on the one balanced-support clip, i.e. chance-level. A co-trained contact head injects it durably (F1 0.227 at full budget, $1.8\times$ a deployable-heuristic baseline at 0.127) at a real fidelity price ($+1.5$ cm at loss weight 0.5), so physics enters the latent only when supervised in.

Control: dominated, not inert. Concatenated beside the explicit reference, the latent is ignored or harmful (-10 pp)—when a more informative channel to the goal exists, the policy takes it. Remove that channel, and the frozen latent alone carries 0.656 ± 0.044 completion through a co-trained prior (§VI). The audit is the specification for §V: the latent becomes an interface only when it is made the sole channel and a prior is co-trained to read it.

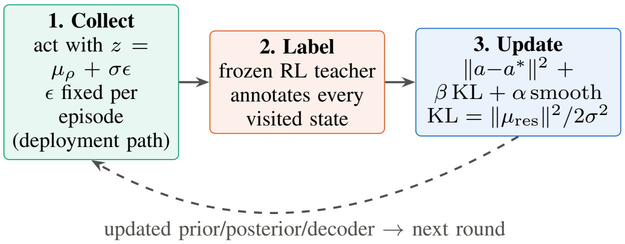

Fig. 2. **One DAgger round (×2,000).** Rollouts are collected on the deployment (prior) path with episodic noise; a frozen explicit-command teacher labels visited states; the CVAE updates with action matching, a closed-form KL from the residual-to-prior posterior (tied σ), and a latent smoothness term. The dashed return edge makes it DAgger rather than behavior cloning. At deployment: $z_{\mathrm{cmd}} = \mu_\rho$ (deterministic), no teacher, no posterior, no explicit reference to the actor.

V. Making the Latent Load-Bearing

The method (Fig. 2) is a conditional VAE trained by online distillation. The **prior** $\rho(z_{\mathrm{cmd}} \mid \mathrm{proprio}, g)$ sees deployable inputs only: proprioception plus the reference goal window (and, in one arm, the frozen SNMR latent stream). The **posterior** adds privileged simulation state as a *residual on the prior mean* with tied fixed σ=0.3, giving the closed-form KL $\|\mu_{\mathrm{res}}\|^2/2\sigma^2$. The **decoder** maps $(\mathrm{proprio}, z_{\mathrm{cmd}})$ to PD targets at 50 Hz and never sees the goal. Training is pure DAgger from a frozen explicit-command RL teacher (no RL gradient mixed in), with episodic reparameterization noise and a latent smoothness regularizer.

Three failures fixed the design, in order. (i) Collecting rollouts on the privileged posterior path collapses deployment to 0.001 while the posterior path itself scores 0.84—train/deploy path consistency is not optional. (ii) Mixing prior-path samples into the action loss damages the posterior (0.84 $\rightarrow$ 0.42) without rescuing deployment. (iii) Goal-conditioning the prior is the load-bearing fix, and a post-repair factorial ablation gives the clean attribution: with a goal-conditioned prior, deployment reaches ≈0.95 whether or not prior-path samples enter the action loss (the mix costs 2.3 pp); without one, 0.000–0.002 either way. And an ablation we ran at reviewers' request completes the attribution: a *deterministic* 64-d goal encoder—no posterior, no KL, no noise, same DAgger loop—reaches 0.969 (one seed, within the CVAE arm's range), and the trained CVAE's own posterior residual is ∼1% of its noise floor. At nominal command rate the variational machinery is not load-bearing; what matters is goal-conditioning, path consistency, and the exclusivity contract. It earns its keep under command-channel *latency*: refreshing z_{cmd} every 100 ms instead of 20 ms, the noise-trained CVAE keeps 0.485 completion where the deterministic encoder collapses to 0.027 (18×; both die at 200 ms). The working design compresses to one line: *a goal-conditioned recoding, trained on the deployment path with injected latent noise, the decoder blind to the goal.*

VI. Experiments

All tracking numbers: 10-second phase-stratified rollouts on a simulated Unitree G1 (MuJoCo-Warp), deterministic deployment path, 1,024 rollouts per evaluation for latent arms (100 for reference policies). *Completion* is the fraction of rollouts that track the full 10 s without triggering the stack's standard tracking-failure termination (anchor drift > 0.5 m or orientation error > 0.8 rad at any tracked body, or end-effector deviation > 0.25 m)—i.e., a rollout "completes" only while it stays within a bounded tube around the reference, not merely upright. Retargeting numbers: 7 held-out LAFAN1 clips, bootstrap CIs over per-clip means. *Recipe:* clip = LAFAN1 walk1_subject5; teacher = PPO, 8k iters, 512 envs, repaired rewards + joint-space term (w=1, σ=0.5); students = 2,000 DAgger rounds, 1,024 envs, 24-step collection, 5 epochs, minibatch 4,096, Adam 3×10^{-4}; prior/posterior 512×256, decoder $512\times256\times128$, β_{KL}=0.1, α_{smooth}=0.005, tied σ=0.3; goal = 58-d reference joints/velocities + 6-d orientation. Code, hashed protocol scripts, and the negative-results ledger accompany the submission.

A. Does the learned interface work?

Table I is the paper's central result: a four-arm factorial over what the prior sees, under one recipe (same DAgger algorithm, reward stack, budget, and seed; 1,024 rollouts per evaluation; all numbers post-repair of both defects in §VII). Completion is monotone in goal quality across completion, RMSE, *and* the imitation loss plateau (0.10/1.03/2.25 for explicit/z_{ret}/no goal). Three readings. First, the interface works: the 64-d command latent as the actor's only motion input *matches* its teacher (0.955 vs 0.954 at equal evaluation precision), where the design's failed variants (posterior-path collection, goal-free priors) collapse to 0.001. An attribution ablation confirms goal-conditioning is the load-bearing design choice: with it, completion is ≈0.95 regardless of the action-loss sampling path (−2.3 pp for prior-path mixing); without it, 0.000–0.002 either way (one seed per cell). Second—and this is the row we were obliged to add—the phase-only clock control decides the z_{ret}-only arm's meaning. That arm retains 0.656 ± 0.044 with no robot-space reference anywhere in the student, is not proprioceptive gait memory (goal-free control 0.001), and is not prior memorization (corrupting the latent collapses it to 0.000, all seeds). But a fixed random projection of frame-index *sinusoids* through the identical fetch path reaches 0.754 ± 0.010—ten points *above* the latent, non-overlapping seed ranges. On a single cyclic clip the deployable content of the retargeting latent is therefore at most timing, delivered more noisily than a clean clock. We had a favorable reading of 0.656 drafted; this control, demanded by our own protocol, retired it. Third, the constructive residue: an exogenous ∼64-d clock alone recovers 79% of teacher completion—goal-conditioned priors need remarkably little information on periodic motion—which is why single-clip results cannot establish latent-content claims (§VIII).

B. Does the latent add value beyond the explicit goal?

On single-clip walking: no—but it substitutes for most of it. The clean factorial separates two questions the leak-era runs conflated. *Additive* value beside the explicit goal is null

TABLE I

COMMAND-INTERFACE FACTORIAL, ONE TRAINING REGIME (G1, WALK1; MEAN±SD OVER 3 TRAINING SEEDS, 1,024 ROLLOUTS PER EVALUATION).

Prior's goal source	Completion	RMSE
— (explicit-command teacher)†	0.954	0.122
Explicit robot-space command	$\mathbf{0.955 \pm 0.008}$	0.127
Explicit cmd + frozen z_{ret}	0.956 ± 0.003	0.127
Phase clock only (control)	0.754 ± 0.010	0.150
Frozen z_{ret} only (human-side)	0.656 ± 0.044	0.161
No goal (proprio-only prior)	0.001 ± 0.001	0.254

†single policy re-evaluated at 1,024 rollouts for precision parity; no seed SD available.

$(0.956 \pm 0.003$ vs 0.955 ± 0.008, paired per-seed deltas straddling zero; the weaker of two redundant channels goes unused, consistent with UniTracker's vanishing-influence observation). *Substitutive* value is large: z_{ret}-only retains 0.656 ± 0.044 where the goal-free control collapses to 0.001. The redundancy variant of the additive claim—the human-side latent should buy robustness when the robot-space reference is corrupted—also fails on its first valid test (the leak-era version noised dimensions the decoder never read): across three seeds and three corruption levels (1,024 rollouts per cell), the paired gap straddles zero at usable noise ($\sigma \leq 0.5$) and reaches only $+1.7$ to $+4.1$ pp where both arms are already broken ($\sigma=1.0$), below the pre-specified $+5$ pp gate—the second sub-gate trend our promotion discipline has correctly rejected. The right reading of the interface program is therefore substitution, not addition: the latent matters where an explicit command is unavailable or unshareable—multi-clip and cross-embodiment settings, where a 29-DoF target stream cannot command a 23-DoF robot at all. Those settings remain out of reach at our compute budget: multi-clip teachers over 8 heterogeneous clips failed their pre-specified quality gate three times (0.38 at 8k iterations, 0.47 at 16k, 0.48 after doubling to 2,048 envs—the registered scaling lever bought $+1.3$ pp, so environment count is not the binding constraint), and we do not distill students from sub-gate teachers. The literature norm for this heterogeneity (4,096 envs, 20–30k iterations) is beyond our budget; we report the plateau rather than claim the setting.

C. *Does the interface scale with better retargeting data?*

Preliminarily, yes (Table II; articulation under ground-truth root—object/terrain state is not yet input, and a redo with full root loss and group-held-out splits is registered). Adding a second, interaction-rich teacher—1,938 OmniRetarget clips [2] whose paired human data arrives on 52- and 53-joint keypoint skeletons, ingested with MST-inferred topology and synthetic orientations—cuts held-out interaction-clip error $5.2\times$ (object) and $3.1\times$ (terrain) while locomotion pays $+1.05$ cm. A sampling-diet ablation attributes 61% of the naive run's larger interference to data balance rather than representational conflict. The dataset also delivers a measured lesson about interaction multimodality: it contains 4,659 sibling pairs with *bit-identical* human motion and divergent robot trajectories (mean DoF divergence 0.031–0.068 rad)—real conditional

TABLE II

TWO-TEACHER ABSORPTION, *preliminary*: HELD-OUT ARTICULATION MPJPE (CM) UNDER GROUND-TRUTH ROOT POSE; MATCHED STEP BUDGET.

Held-out pool	LAFAN1	Two-tchr	Δ
Locomotion (LAFAN1)	**3.31**	4.36	$+1.05$
Object inter.	28.95	**5.53**	-23.4 ($5.2\times$)
Terrain inter.	25.06	**8.16**	-16.9 ($3.1\times$)

multimodality, 16–36× the threshold below which our earlier work showed generative decoders have nothing to model. We tested whether conditioning explains it: a single terrain-scale scalar recovers 52% of terrain-sibling variance, but the exact per-frame 7-d object pose recovers only 6% of object-sibling variance under the same instrument. Terrain multimodality is hidden-variable-explained; object multimodality (grasp-strategy choice, with arm excursions up to 4.7 rad between siblings) is not, so a deterministic decoder averages modes on exactly the interaction-rich clips—the measured justification for a generative decoding head, which we are testing under a pre-specified kill rule.

D. *Are learned references as trackable as IK references?*

Matched tracking policies (3 seeds per source × 2 eval seeds × 100 rollouts) trained on SNMR references vs the IK teacher's: 86.7% vs 86.7% completion—statistically indistinguishable (per-seed spread ±3 pp; the exact tie is coincidence at this power)—with joint RMSE noninferior (+0.33%, CI $[-3.1, +4.9]$%). This matrix predates the reward repair (§VII); both sources trained under the identical defect. A sensitivity control now validates the assay itself: a tracker trained on a deliberately corrupted reference (i.i.d. joint noise, $\sigma=0.05$ rad — the scale of known retargeting artifacts) drops to 0.48 completion against its own reference, a 46-point fall that is 9× our pre-specified detection gate. The completion assay is therefore sensitive to reference-quality differences of the relevant magnitude, and the observed point-equality is informative rather than an instrument blind spot. (The control's corruption is i.i.d. while real retargeting error is spatially structured; a post-repair rerun on harder clips remains registered.) Formal noninferiority on completion was not established: the ±6 pp CI exceeds the pre-specified −5 pp margin at this seed count—a power statement, not a quality gap (per-seed spread ±3 pp dominates any source effect).

E. *What does physics-repaired supervision buy and cost?*

Rolling a calibrated tracker out on its own references and recording simulated states yields repaired supervision with 2–3× lower stance-foot speed than the references themselves (3–6× on pre-repair policies) and zero penetration (a simulator property, which we do not claim), at 6.2–6.7 cm heading-local reconstruction distance—the quantified data-side alternative to bilevel physics retargeting [1].

VII. A Defect in the Measurement Substrate

While calibrating our tracker against external baselines we found that the primary body-position tracking reward had been *exactly zero* for every one of our MuJoCo-Warp runs: a world-body off-by-one (state tensors include the world body at index 0; the body-name list excludes it) shifted every simulation-side body read one body rootward in our stack (pinned revision; IsaacSim/IsaacGym paths unaffected), so the tracked pelvis slot read the world body—always at the origin—and the exponential reward kernel underflowed to 0.0 for 8,000 consecutive iterations (the raw term logs show it). Policies still walked, because orientation and velocity terms survived; they were orientation-plus-velocity trackers with a corrupted position channel. Repairing the indexing (released as a small patch; IsaacSim/IsaacGym paths are unaffected, which is why upstream never saw it) cut joint tracking RMSE from 0.263 to $0.142\,\mathrm{rad}$ (-46%) with no change to the learning algorithm; adding a standard joint-space reward term then reached $0.122\,\mathrm{rad}$ at 0.98 completion, matching an external 187-run calibration band (97.9%). Paired comparisons made under the defect survive (both sources suffered identically); absolute capability numbers did not. We report this at section level because the lesson generalizes: *measure your reward channels before tuning them.*

The same discipline then caught a second defect—in our own code. The observation manager of our tracking stack concatenates observation terms *alphabetically*, not in configuration order; our distillation trainer sliced the explicit reference off the front of the observation vector, where it is not. The consequence, established by causal ablation (blanking the leaked dimensions at deployment drops completion from 0.93 to 0.00), is that in the affected runs the decoder saw the explicit reference inside its "proprioception" input, violating the exclusivity contract of §V. All interface-side numbers in this paper are from post-repair reruns; retargeting-side results, the audit probes, the reference-policy baselines, and the trackability matrix were never affected (their pipelines do not slice the observation vector). The repair moved the results in *both* directions: the explicit-goal arm improved (0.935 to 0.955 ± 0.008—the corrupted goal slice had been hurting its prior), and the frozen-latent-only arm went from 0.004–0.016 to 0.656 ± 0.044—under the leak, the always-on explicit shortcut inside "proprio" suppressed learning from the latent entirely; a false negative we had logged reopened—then was cut back down by the clock control (§VI), each step gated in advance. The red flag that should have caught it earlier: under the leak, arms with *identical* goal information differed by ninety points of completion—an incoherence input duplication cannot produce.

VIII. Limitations

Interface scope — the central open step. On single-clip cyclic walking the frozen retargeting latent's deployable value is bounded by the clock control (0.656 vs 0.754): no latent-content claim survives on this instrument. The live forms of the question are (i) aperiodic/multi-clip tracking, where phase alone cannot suffice—blocked at our budget by the teacher quality gate; (ii) cross-embodiment command, where explicit references are unshareable by construction; and (iii) tracking gradients into the retargeting encoder. The latent's additive value beside an explicit goal is null on single-clip walking. **Embodiment generalization.** Zero-shot transfer to an unseen robot fails ($5.2\times$ error); embodiment augmentation is the identified path. **Physics of the latent.** Contact enters only by co-training at a fidelity price; repaired references cost 6+ cm at current tracker quality. **Interaction tracking.** Our simulator path supports neither objects nor terrain in the tracking loop; interaction data enters at the retargeting level only, where its absorption is measured. **Simulation only.** No hardware; the MuJoCo-Warp stack is externally calibrated but sim-to-real is unvalidated. **Data scale.** 4.6 h/robot of locomotion plus 0.9–3.4 h of interaction data is small; the two-teacher result is the first point on the scaling curve, not its end. **Interaction multi-modality is not yet modeled.** Conditioning on the interaction hidden variable fails even given the exact per-frame object pose (6% of object-sibling variance recovered, vs. 52% for terrain from a single scalar)—deterministic decoding averages modes on precisely the interaction-rich clips; a generative head with a pre-specified kill rule is under test. Moving past these—closing the encoder-gradient gap, capacity scaling matched to data, and the cross-embodiment command demonstration—is the immediate agenda.

IX. Conclusion

The boundary between retargeting and tracking has been a file format. Treated as a representation—audited before trusted, co-trained before deployed, and made the policy's only channel to the goal—the learned code matches the explicit interface it replaces (0.955 vs 0.954). The retargeting latent itself did not survive its controls: what looked like two-thirds of teacher completion from human motion alone is bounded above by a trivial clock (0.754), and we publish that bound as the null model the next act-through-latent claim must beat. The measurement lesson runs deeper than the architecture: this paper's strongest results include a reward channel that was silently dead, a set of pre-specified nulls, a false negative overturned by a defect repair, and a false positive retired by a pre-specified control—we commend all four practices, and especially the last, to the field.

References

[1] D. Müller et al., "ReActor: Reinforcement learning for physics-aware motion retargeting," *ACM TOG*, 2026.

[2] L. Yang et al., "OmniRetarget: Interaction-preserving data generation for humanoid whole-body loco-manipulation and scene interaction," arXiv:2509.26633, 2025.

[3] J. P. Araujo et al., "Retargeting matters: General motion retargeting for humanoid motion tracking," arXiv:2510.02252, 2025.

[4] Q. Liao et al., "BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion," arXiv:2508.08241, 2025.

[5] Z. Chen et al., "GMT: General motion tracking for humanoid whole-body control," arXiv:2506.14770, 2025.

[6] K. Yin et al., "UniTracker: Learning universal whole-body motion tracker for humanoid robots," arXiv:2507.07356, 2025.

[7] Z. Luo et al., "Universal humanoid motion representations for physics-based control," *ICLR*, 2024.

[8] C. Tessler et al., “MaskedMimic: Unified physics-based character control through masked motion inpainting,” *SIGGRAPH Asia*, 2024.
[9] H. Yao et al., “ControlVAE: Model-based learning of generative controllers for physics-based characters,” *SIGGRAPH Asia*, 2022.
[10] T. He et al., “HOVER: Versatile neural whole-body controller for humanoid robots,” *ICRA*, 2025.
[11] M. Ji et al., “ExBody2: Advanced expressive humanoid whole-body control,” arXiv:2412.13196, 2024.
[12] X. Chen et al., “Implicit kinodynamic motion retargeting for human-to-humanoid imitation learning,” arXiv:2509.15443, 2025.
[13] W. Zhao et al., “Make tracking easy: Neural motion retargeting for humanoid robots,” arXiv:2603.22201, 2026.
[14] H. Zhang et al., “AdaMorph: Unified motion retargeting via embodiment-aware adaptive transformers,” arXiv:2601.07284, 2026.
[15] K. Yin et al., “Behavior foundation model for humanoid robots,” arXiv:2509.13780, 2025.
[16] Y. Li et al., “BFM-Zero: A promptable behavioral foundation model for humanoid control using unsupervised reinforcement learning,” arXiv:2511.04131, 2025.